Presentation Video — narration, v3

iGEM Claremont-CA 2026 | draft of 2026-10-24

Timecodes assume 140 words per minute; the uploaded file has a hard limit of 15:00.

Line numbers run continuously — send edits back as *line 212: …*

Opening

120 words ends 0:51

ON SCREEN — Team logo, a team photo, then the title card.

We're iGEM Claremont-CA, a first-year team in California, and this is Phagenomics.

Pseudomonas causes serious infections in immunocompromised patients and persistent crop disease in agriculture, and phage therapy is one of the alternatives to antibiotics people are betting on. Its bottleneck is host range: each phage binds a specific receptor on a specific strain, and we can't design that specificity yet, because we don't understand how a phage recognises its target in the first place.

So we didn't set out to build a therapeutic. We set out to understand the recognition step any therapeutic would depend on. We went at it four times this season. Three of those attempts failed, and the failures turned out to be the useful part.

Act I.1 | The bottleneck

62 words ends 1:18

ON SCREEN — Plaque assay animation; then the same drop on two other strains, no effect.

A drop of phage goes onto a lawn of bacteria, and a few hours later there's

a clear circle where the bacteria used to be. Move the same drop to the next plate and nothing happens.

Same phage, different strain. That fussiness is the bottleneck: you need a phage that matches the strain in front of you, and usually there isn't one.

Act I.2 | Adsorption and the sugar coat 63 words ends 1:45

ON SCREEN — Cross-section of a cell: membrane, wall, O-antigen, capsule. Mark which layer the phage meets first.

It gets decided at the first step, before any DNA moves: the phage has to recognise the outside of the cell and stick to it.

And many bacteria aren't presenting a bare protein surface. They're wrapped in sugar — long chains called the O-antigen, and above that, in some species, a capsule. The first thing a phage touches is a forest of sugar.

Act I.3 | Where the project came from 61 words ends 2:11

ON SCREEN — Team photo with Prof. Libeskind-Hadas, captioned as an early conversation.

One thing belongs at the start, because it's why half this project exists.

Early on, Professor Ran Libeskind-Hadas at Claremont McKenna suggested reconstructing an ancestral phage — a whole organism, inferred from its relatives. We couldn't build one. But scale that down to a single enzyme and you can order it from a synthesis company. That is what this

project became.

Act I.4 | What we isolated

138 words ends 3:10

ON SCREEN — Sampling photo; five isolates collapsing to three strains; the circular phage genome with the tail spike outlined.

Soil from the California Botanic Garden, on the nineteenth of May. Out of it came five *Pseudomonas* isolates, whole-genome sequenced, all five closing into complete circular genomes. They collapse into three strains that differ from one another — two of them the same species, the third a different one.

That leaves two pairs of near-identical isolates, and those pairs do real work for the rest of the project. They're our internal control: any difference we report between strains has to be bigger than the difference we can measure within one.

And one phage, Phi-8, which infects exactly one of the three strains. Sixty-two thousand, three hundred and thirty-nine base pairs, circular, eighty genes, no match in the phage database, and effectively no genes in common with T7 or anything else characterised. The architecture is familiar. The sequence isn't.

Act I.5 | The first clue

65 words ends 3:38

ON SCREEN — The three O-antigen loci side by side, labelled as a hypothesis.

Then we lined the three strains up at the locus that builds their sugar

coat. Same position, same flanking gene — and inside, three cassettes with no homology to each other at all.

Three strains, three different sugar coats, one phage that kills one of them. So our first hypothesis: the sugar coat might be one of the main things setting what Phi-8 can attack.

Cycle 1 | Design

74 words ends 4:09

ON SCREEN — DBTL bar, DESIGN lit. Lock and key.

But that isn't the hypothesis we tested first.

The textbook picture of attachment is a receptor-binding protein on the tail meeting a protein on the bacterial surface. Lock and key. If that's what you assume, the obvious move is to work out which binding protein fits which receptor.

If we'd known the receptor was a sugar we'd never have run this. We thought it was a protein, so we went looking for a protein.

Cycle 1 | Build

181 words ends 5:27

ON SCREEN — BUILD lit. Funnel from 441 to 144 clusters to 114 × 63. The surrogate-host trade-off as two columns. Three tools and why each stalled.

First we needed to know what was actually on the surface. We annotated four hundred and forty-one candidate surface receptors across all five isolates and collapsed them into a hundred and forty-four clusters.

Then we needed a reference set of phages, which meant choosing a sur-

rogate host — a relative of our strains that people have already characterised phages on. The closest match by receptor content had only twelve phages on record. A slightly more distant relative had two hundred and twenty-seven binding-protein entries across seventy-one phages. We took the one with the literature rather than the one with the best identity, because a receptor shared with a strain nobody has studied tells you nothing.

That gave us a hundred and fourteen binding proteins against sixty-three receptors — seven thousand, one hundred and eighty-two pairs — each folded as a complex on our university's GPU cluster and ranked by the interface score.

Getting it to run took three tools: ColabFold throttled by its alignment server, Boltz-2 blocked by the cluster's CUDA version, and finally ESMFold, which was the one that ran.

Cycle 1 | Test

136 words ends 6:25

ON SCREEN — TEST lit. The ranking first so it looks reasonable, then score against protein length, then the control figure. Hold 0.137 and 0.136 large.

Six thousand, three hundred and fifty-eight pairs came back with a score. And the ranking looked fine, which is the trouble with this kind of output — it always looks fine.

What made us uneasy was that the scores tracked protein length. Correlation of plus nought point six.

So before building anything on it, we ran the method against pairs where

the answer was already known: complexes solved experimentally, against the same receptors shuffled into nonsense, and against deliberately wrong pairings.

T5's pb-five with FhuA, a real co-crystallised complex, scored zero point one three seven. The same receptor shuffled scored zero point one three six.

We assumed that was the tool, so we rebuilt it with real evolutionary alignments. It got worse: a deliberately mismatched pair came out at zero point five, above every genuine binder.

Cycle 1 | Learn

220 words ends 8:00

ON SCREEN — LEARN lit. Two confidence bars: on scrambling the left collapses 0.55 to 0.17, the right does not move. Then two circles that never intersect.

Two tools, two control gates, both failed. One failure is about your run; two independent failures are about the method.

These models report two confidences: one for the shape of the molecule, one for how two molecules sit together. When we shuffled a sequence, the shape confidence collapsed from zero point five five to zero point one seven — the model could tell it was nonsense. But the interface number barely moved. It had never been reading an interface at all.

The reason is that these models don't simulate two proteins touching. They read evolution — looking across thousands of species for positions that change in step. If a residue on one side mutates and one on the other mutates to match, those two are probably in contact.

That only exists if the two proteins have been inherited together, in one genome, for a long time. A phage protein's relatives are in other phages; a bacterial protein's are in other bacteria. They're never in the same genome, so the record of that interface doesn't exist. Not hidden — never written.

So this was never a result about our proteins. It's a property of the method for any phage-and-host pair, and more compute doesn't help. Nine control folds decided the fate of seven thousand, and we never ran the real screen.

Cycle 2 | Design and Build

55 words ends 8:23

ON SCREEN — An open-source classifier over 80 proteins, 8 candidates, rung-period diagram.

If we couldn't ask what the tail spike binds, we could ask what it is.

We ran an open-source classifier across all eighty Phi-8 proteins to pick the candidates, folded the eight that came out, and classified their folds by the spacing of the backbone hydrogen bonds — which needs no structure database at all.

Cycle 2 | Test and Learn

192 words ends 9:45

ON SCREEN — Fold table: eight proteins, four verdicts, so the method discriminates. Then the beta-helix render and a sugar-cutting animation.

Eight proteins, four different verdicts — so the method discriminates rather than just agreeing with us. It even called one of our own targets

not a beta-helix, which matched an independent read of that protein as a completely different fold.

CDS 11, the tail spike, came back a right-handed beta-helix: a long triangular prism, classifier confidence nought point nine nine seven, fold confidence ninety-three. And CDS 24, which the annotation had written off as a hypothetical protein, came back as a second depolymerase the homology tools had missed.

That fold is a signature. Enzymes that chew through long sugar chains look like this, because a sugar chain is a thread and you need something that holds it straight while you cut. You can recognise a wood plane by its shape without being told what it cuts.

So the tail spike isn't a key. It's a blade — and the lock we'd spent months looking for doesn't exist.

That reading arrived from two directions at once: our classifier flagged it as a depolymerase, and our PI reached the same conclusion from his own structural search. Between us we turned it into a hypothesis.

Cycle 2 | The hypothesis

68 words ends 10:15

ON SCREEN — One-sentence hypothesis box on white, held. Credited jointly with the PI.

In bacteria that wrap themselves in sugar, the ceiling on host range is set by what the depolymerase can digest — not by which protein the tail binds. And the narrow specificity we see now is a narrowed-down version of something broader further back.

That tells you what to measure — substrate range, not binding — and where to look for the broader version: back up the tree.

Cycle 3 | Design

147 words ends 11:18

ON SCREEN — Domain map of CDS 11: anchor, flexible shaft, rigid body. Then the tree collapsing to three nodes.

If the depolymerase is the thing that matters, the test is whether that domain, on its own, can cut the sugar.

CDS 11 is nine hundred and forty-one residues, and most of it isn't the enzyme. The first hundred and ten anchor it to the virion. The next two hundred are a shaft that moves between every model we ran. The rigid trimeric body — residues three-oh-one to nine-four-one, agreeing between models to within one ångström — is the part that carries the active site. That's what we submitted, and what we expressed.

Then the ancestors, which is where that first conversation comes back. We worked out the direction with our PI; he ran the alignment, the tree and the reconstruction on sequences we prepared, and the further back up the tree he went, the better a protein language model scored the result. We picked three nodes.

Cycle 3 | Build

134 words ends 12:15

ON SCREEN — Registry cards for the six parts, published and CC0. Caption: three of the six are sequences that exist nowhere in nature.

Six constructs, all in pET-28a with an N-terminal His-tag, synthesised by

Twist across two orders — and all six are now published basic parts on the iGEM Registry, under a CC0 licence.

Three of those six are the reconstructed ancestors: protein sequences that exist nowhere in nature, inferred from an alignment and a tree and then ordered as DNA. That route to new parts is reusable by any team working on an enzyme family, so we wrote the method out in full.

One warning we'd pass on, because it cost us. Our first Twist order was codon-optimised and our second wasn't — the second set carries a mammalian-like codon bias, which in an *E. coli* host is the first thing to suspect when a band doesn't appear. Anyone reusing those sequences should re-optimise first.

Cycle 3 | Test

134 words ends 13:12

ON SCREEN — Four test rounds. The 9-15 gels, the 9-29 band, the 10-01 elution that worked, the 10-10 column that did not.

Four rounds at the bench, and they didn't line up neatly.

September the fifteenth: two gels, four constructs, induced and uninduced — no detectable induction on either. Inconclusive. The tail spike lane was essentially blank, which points at lysis or loading rather than at expression.

September the twenty-ninth: CDS 12 gave the first induction-dependent band at the right size.

October the first: we ran nickel columns on six constructs, and one worked. The first ancestor eluted at its predicted mass — the only con-

struct that did. The other five gave nothing.

October the tenth: the same column on a different ancestor, and both elutions came off blank, while the load and the flow-through carried the same band. We read that as a purification we'd botched, and were about to spend another week on it.

Cycle 3 | Learn: an outside view changed a result

240 words ends 14:55

ON SCREEN — The strongest human-practices beat in the video. Tolstorukov's email, then the 10-19 dot blot with 2.9× marked.

What stopped us was a conversation.

On October the thirteenth we took it to Professor Ilya Tolstorukov at Keck Graduate Institute, who works on protein expression. He wrote back before we met to say he wasn't sure he was the right person — expression is his field, purification isn't — and met us anyway. That turned out to be the point.

He changed the question. Not *why is nothing eluting*, but *is there anything there to elute*. Check whether the protein is being made at all — a dot blot against an anti-His antibody — before touching the column again. He also told us to run uninduced controls, and to standardise how much material goes onto a membrane before comparing spots on it.

Six days later we ran that blot. The flow-through from the second ancestor read two point nine times the negative control.

So the protein was there. It had been made, it came off in the flow-

through, and the run failed at capture — not at expression. A gel can't tell those apart: no expression, insoluble protein, and a column that didn't bind all look identical. One cheap assay separates them, and it pointed the fix at the column instead of the construct.

We'd spent a week debugging the last step of a workflow without checking the first step worked, because the last step was the one visibly failing. That's the generalisable mistake, and one meeting fixed it.

Cycle 3 | The activity assay

119 words ends 15:46

ON SCREEN — The 10-16 spot test and colony counts normalised to the PBS control. Caption the missing carry-over control.

And then the assay the whole project points at: does this enzyme, on its own, do anything to the sugar?

We read it on October the sixteenth. No visible lysis on any plate — including Phi-8 on its own hosts, which tells you the assay wasn't reading clearing. But counting colonies inside the drops gives a number. Under the first ancestor, colony counts fall to a median of twelve percent of the plate's PBS control. Phi-8's own median is zero.

Our PI reads that suppression as enzyme activity. We'd want one more control before we call it that ourselves — the one that separates protein activity from phage carried over into the preparation — and that control wasn't run.

Cycle 3 | What we still owe

49 words ends 16:07

ON SCREEN — Honest list.

And the gap that runs through all of it: we have now run eight protein assays — seven gels and a dot blot — and not one carries a lane of untransformed BL21. Every round was designed around a different variable instead. That control is on the bench now.

Cycle 4 | Design and Test

99 words ends 16:50

ON SCREEN — Colonies from the plaque zone, whole-genome sequencing, read-level verification, F126L inside a 30 kb capsule cluster.

Three cycles in, every guess about the receptor had come from the phage's side. So for the fourth we let the bacteria answer.

In July we picked colonies from inside the plaque zone — cells still growing where the phage should have killed them — and sequenced them against their parents. No structure prediction anywhere in this loop.

Every difference was checked back against the raw reads, which is the step that removed our own false positives. Exactly one mutation survived: a single amino-acid change in a sugar phosphotransferase, inside a thirty-kilobase capsule cluster. The O-antigen cassette was untouched.

Cycle 4 | Learn

108 words ends 17:36

ON SCREEN — A capsule hiding an intact O-antigen: how both conclusions hold at once.

That disagrees with Cycle 2, which is the whole reason for doing it this

way round. Cycle 2 said the receptor is a sugar. Cycle 4 agrees it's a sugar and disagrees about which one — it points at the capsule, not the O-antigen underneath.

Both can be true at once, because a capsule can hide an intact O-antigen underneath it. We don't know yet which surface Phi-8 reads, so where the distinction matters this project says *surface polysaccharide* and leaves it there.

And the honest gap: those colonies were never confirmed resistant. They were colonies in a plaque zone, which is suggestive and isn't the same thing.

What we're leaving behind

160 words ends 18:45

ON SCREEN — Tool terminal output, the GitLab repository, the six parts, the negative result.

Four things, and one of them is a failure.

Two packaged tools, MIT licensed, on iGEM's GitLab. One does ladder calibration and densitometry on a gel photograph — we re-typed that code three times across three gel days before admitting we should generalise it. The other tells a direct terminal repeat from a circular join by the shape of the read coverage, and that one changed our own answer: this wiki used to say the genome was linear, the check said circular, and circular is what we now report. Two more tools are staged with porting notes.

Six parts on the Registry, three of them sequences that never existed.

A benchmark: a machine-learning classifier found a depolymerase on a novel genome that homology-based annotation had labelled hypothetical.

And the screen that failed — published with the controls that caught it and the reason it could never have worked, so the next team planning one can read it and skip it.

Coda | The garden

209 words ends 20:14

ON SCREEN — Interview clips, at least 20 seconds. Credit both hosts and note permission.

Two days before our wiki froze we went back to the garden the soil came from, and asked whether a treatment as strain-specific as ours would be any use to them.

Most of what we'd assumed was wrong. We expected to argue for biological control against a chemical default, and didn't have to — they already suppress soil fungi by mulching. We expected the objection to be ecological risk. It wasn't; it was whether the thing works. Copper is the only chemistry they have for bacterial disease, it goes onto every blossom every four or five days, and they have eighty-six acres and eight staff. The objection is arithmetic.

And the finding that costs us most: they've never had a sample come back positive for *Pseudomonas*. We isolated a soil phage and its environmental host — not a treatment for a disease this stakeholder has.

What they do need is something for bacteria inside the vascular system,

where nothing they have reaches. Our strain doesn't solve that. But the question we worked on — what a tail spike reads on a cell surface, and how far that can be predicted from a genome — is the one you'd have to answer first. The strain is local. The question isn't.

Coda | Safety

121 words ends 21:06

ON SCREEN — The IBC letter, 0 VFDB / 0 CARD, and the open condition quoted.

On safety: our institutional biosafety committee gave conditional approval with one condition still open, and that condition is a question about host range — the subject of this entire project. They asked us to show our reasoning that the work won't alter host range, virulence, or antibiotic resistance. We don't have full approval; the revision is being written.

Phi-8 carries no virulence factors and no resistance genes, and our constructs are protein fragments in lab *E. coli* — nothing here is a modified phage. But understanding what sets the ceiling on host range is the same as knowing what would have to change to raise it, so we treat this as dual-use rather than arguing our way to a clean no.

Coda | Where we are

199 words ends 22:31

ON SCREEN — Three columns fading in: Established / Partly tested / Open.

So, where we actually are.

Established: five bacterial genomes and a novel circular phage genome;

three unrelated sugar cassettes in one locus; a tail spike that is a sugar-cutting enzyme by two independent methods; a controlled negative result about interface prediction; and one resistance mutation verified in the reads.

Partly tested: that Phi-8 won't infect the other two strains — a host-range matrix in July pointed the right way, but carried no no-phage control. And the enzyme itself: one ancestor purified at its predicted mass, and colony counts under it fall to twelve percent of control, which our PI reads as activity and we won't call that ourselves without the carry-over control.

Open: a clean expression round with the control we keep not running; which sugar layer the phage reads; whether Phi-8 is temperate; and depositing the genome, which we haven't done.

Four cycles, three of them wrong, each wrong in a way that told us where to go next. If there's one thing we'd pass on: put the cheap control in front of the expensive experiment. Nine folds told us what seven thousand couldn't, and one afternoon's dot blot told us what a week on a column hadn't.

3154 spoken words. With the act cards and a 30-second credit card, the finished file runs about 23:31 at 140 words per minute.